

3. (2 points)

Given the following information as of December 31, 2017:

AY	Earned Premium (\$000)	Cumulative Paid Loss (\$000) as of (months)			
		12	24	36	48
2014	8,000	2,500	3,335	3,942	4,021
2015	8,320	2,100	2,705	3,335	
2016	8,650	3,000	4,113		
2017	9,000	3,500			

- Assume there is no further development after 48 months.
- $t_i = \sqrt{p_i}$
- $E[\alpha_2^2 (U_2)] = 2,000$

Calculate the mean squared error for both the individual loss ratio method and the collective loss ratio method, and determine which is preferable for estimating R_{2015} .

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 3

TOTAL POINT VALUE: 2

LEARNING OBJECTIVE(S): A1

SAMPLE ANSWERS

Sample 1

$$\text{MSE} = 2,000 \times [z^2/p + 1/q + (1-z)^2/\sqrt{p}] \times q^2$$

Individual: $Z = 1$

Collective: $Z = 0$

MSE Ind = 38.736

MSE Coll = 38.729

Collective is preferred since the MSE is lower.

$$m_1 = (2.5 + 2.1 + 3.0 + 3.5) / (8 + 8.32 + 8.65 + 9) = x = 0.327$$

$$m_2 = (3.335 - 2.5 + 2.705 - 2.1 + 4.113 - 3.0) / (8 + 8.32 + 8.65) = y = 0.102$$

$$m_3 = (3.942 - 3.335 + 3.335 - 2.705) / (8 + 8.32) = z = 0.076$$

$$m_4 = (4.021 - 3.942) / 8 = t = 0.0099$$

$$x + y + z + t = a = \text{ELR} = 0.515$$

$$p_{2015} = (x + y + z) / a = 0.981$$

$$q_{2015} = 1 - p_{2015} = 0.019$$

Sample 2

$$\text{MSE Formula: } E[\alpha^2(U)] \times q^2 \times [z^2/p + 1/q + (1-z)^2/t]$$

Incremental Paid Loss

AY	12	24	36	48
2014	2,500	835	607	79
2015	2,100	605	630	
2016	3,000	1,113		
2017	3,500			

$$m_{12} = (2,500 + 2,100 + 3,000 + 3,500) / (8,000 + 8,320 + 8,650 + 9,000) = 0.33$$

$$m_{24} = (835 + 605 + 1,113) / (8,000 + 8,320 + 8,650) = 0.10$$

$$m_{36} = (607 + 630) / (8,000 + 8,320) = 0.08$$

$$m_{48} = 79 / 8,000 = 0.01$$

$$m_{\text{total}} = 0.33 + 0.10 + 0.08 + 0.01 = 0.51$$

$$p_{36} = (0.33 + 0.10 + 0.08) / 0.51 = 0.98$$

$$q_{36} = 1 - p_{36} = 0.02$$

$$t_{36} = \sqrt{0.98} = 0.99$$

For individual LR method, $Z = 100\%$, $\text{MSE} = 2,000 \times (1 - 0.98)^2 \times [1^2 / 0.98 + 1 / (1 - 0.98) + 0] = 41$

For collective LR method, $Z = 0\%$, $\text{MSE} = 2,000 \times (1 - 0.98)^2 \times [0 + 1 / (1 - 0.98) + 1 / 0.99] = 41$

Since the MSEs are roughly equal, they are equally preferable.

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample 3

$$\text{MSE} = E[\alpha^2(U)] \times q^2 \times [z^2/p + 1/q + (1 - z)^2 / t]$$

$$m_1 = (2,500 + 2,100 + 3,000 + 3,500) / (8,000 + 8,320 + 8,650 + 9,000) = 0.327$$

$$m_2 = (835 + 605 + 1,113) / (8,000 + 8,320 + 8,650) = 0.102$$

$$m_3 = (607 + 630) / (8,000 + 8,320) = 0.076$$

$$m_4 = 79 / 8,000 = 0.01$$

$$m_{\text{total}} = 0.327 + 0.102 + 0.076 + 0.01 = 0.515$$

$$p_2 = (0.327 + 0.102 + 0.076) / 0.51 = 0.981$$

$$q_2 = 1 - 0.981 = 0.019$$

$$t_2 = \sqrt{0.981} = 0.990$$

$$Z_2 \text{ Individual} = 1$$

$$Z_2 \text{ Collective} = 0$$

$$\text{MSE}(\text{Ind}) = 2,000 \times [1/0.981 + 1/0.019 + 0] \times 0.019^2 = 38.74$$

$$\text{MSE}(\text{Coll}) = 2,000 \times [0 + 1/0.019 + 1/0.99] \times 0.019^2 = 38.73$$

$$Z_2 \text{ Optimal} = p_2 / (p_2 + t) = 0.498$$

Since Z Optimal is closer to 0 it assigns more weight to the Collective method. Therefore, the Collective method is preferable to estimate Z_{2015} .

EXAMINER'S REPORT

Candidates were expected to calculate the inputs needed to determine the mean squared error for the individual loss ratio method and collective loss ratio method.

Specifically, candidates were expected to determine the following:

- Incremental amount of expected paid claims per unit of exposure for each development period.
- Loss ratio payout factor and loss reserve factor.
- The amount of credibility for each method.

Common mistakes included:

- Not knowing the mean squared error formula for the two methods.
- Using the loss ratio payout factor and loss reserve factor from the wrong period.
- Using a method other than the loss ratio method to calculate the payout factor and reserve factor.
- Using the wrong amount of credibility for each method.